

work (the jargon for this is the *participation constraint*). The contract must also provide the right incentives for these agents to work hard, lest the talented fund managers spend their time playing golf (this is called the *incentive compatibility constraint*). The contract maximizes the principal's value, and the contract takes into account the participation and incentive compatibility constraints.

Often the talented fund manager will be able to generate value that the asset owner cannot. This surplus will be split between the fund manager and the asset owner in a *bargaining game*. In some cases, the fund manager may extract all the value, leaving the asset owner with nothing. A cynic would say the technical term for this is *Wall Street*. Fred Schwed's famous book, *Where Are the Customer's Yachts?*, is as relevant today as it was when it was published in 1940:

Once in the dear dead days beyond recall, an out-of-town visitor was being shown the wonders of the New York financial district. When the party arrived at the Battery, one of his guides indicated some handsome ships riding at anchor. He said, "Look, those are the bankers' and brokers' yachts."

"Where are the customers' yachts?" asked the naïve visitor.

I will show over the next few chapters that in several investment vehicles—mutual funds, hedge funds, and private equity—the average fund manager adds zero to, or subtracts value from, an asset owner using an appropriate benchmark. That is, the fund manager collects, on average, all the surplus. In some cases, however, the fund manager may still add value because the asset owner may not be able to implement the benchmark herself. In other cases, the client wants someone to hold his hand and the fund manager helps him stay the course (see chapter 4).

2.2. OPTIMAL CONTRACTS IN GENERAL AGENCY PROBLEMS

Solving principal-agent problems and finding optimal contracts is difficult. Two Nobel Prizes have been awarded in this area: James Mirrlees received one in 1996 for solving models involving both adverse selection and moral hazard. (He is best known for his result, cited by the Nobel committee, that the optimal marginal tax rate for the rich should be 0%.) Leonid Hurwicz, Eric Maskin, and Roger Myerson were awarded the 2007 Nobel for developing the field of mechanism design, an area closely linked with agency theory. Many principal-agent problems can be interpreted as games and solved using the insights developed by these authors.⁵

⁵Textbook treatments of this material are Salanie (1997) and Bolton and Dewatripont (2005).